World Wide Technology

Stop Dreaming, Start Engineering

Cloud Patterns for Production AI

Jeremy Meiss — Tech Solution Architect

"

How many of you have shipped an AI demo, feature, or new application that quietly died?

80%+

of AI projects fail — twice the rate of conventional IT

RAND, 2024. 65 practitioner interviews — citing an outside estimate.

Two more studies

95%

of GenAI pilots show no measurable P&L return

The GenAI Divide. MIT Project NANDA, 2025.
Preliminary, not peer-reviewed.

28%

of AI use cases in infrastructure & operations meet ROI expectations

Gartner, April 2026. 782 I&O leaders.



Jeremy Meiss

Tech Solution Architect · WWT

- DevOpsDays KC organizer
- CommunityDays KC organizer
- CDF Ambassador 2025-26

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Other Awesome Organizations







Where the failure actually is

The notebook

model.fit() converges

Accuracy looks great on the holdout set

Demo day: it works

The system around it

Data pipeline — unowned, undocumented

Identity boundary — nobody drew it

Observability — bolted on after the incident

Cost line — nobody modeled it

The story we tell ourselves vs. the reality

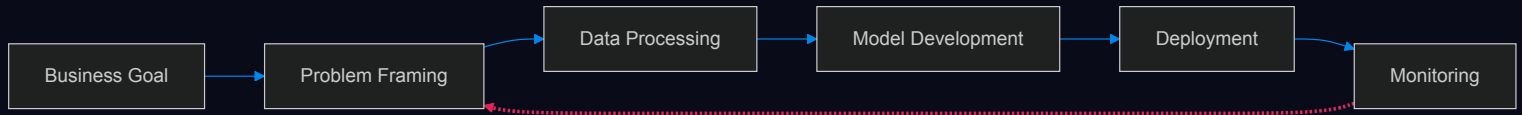
The story...	The reality...
model works in the notebook! we're 90% done!!!!!	When the model works in the notebook, you're maybe 10% done.

The other 90% is the engineering we've spent 30 years figuring out for every other kind of software + some nrew rules for the parts that are different.

"

AI is software with new rules.

The six-phase lifecycle



Use-case drift

Week 1

"Reduce complaints by 30%"

Month 3

Which vector database?

Month 6

Nobody mentions complaints anymore

One of RAND's five root causes: stakeholders miscommunicating the problem

Well-Architected, for AI

Operational Excellence	Security	Reliability
MLOps automation	Model, data, prompt	Recovery from drift

Performance	Cost	Sustainability
Purpose-built silicon	Inference is your bill	Measurable now

AWS ML Lens · GCP AI/ML Framework · Azure Well-Architected AI workload guidance

Three competitors. Same six answers.

AWS **GCP** **Azure**

All three arrived at the same six Well-Architected pillars — independently.

AWS ML Lens · GCP AI/ML Framework · Azure Well-Architected AI workload guidance

"

Regulation is an architectural input now. Not a Phase-6 checkbox.

The EU AI Act. Two dates, not one.

August 2, 2026

- Article 50 transparency obligations
- GPAI penalty powers
- Fines up to €15M or 3% of global revenue

Enforceable 5 weeks ago

December 2, 2027

- High-risk regime
- Human-in-the-loop checkpoints
- Risk classification, conformity assessment

Delayed 16 months — Digital Omnibus, May 2026

Source: EU AI Act (Regulation (EU) 2024/1689); Digital Omnibus package, May 2026

Standards, not just regulations

EU AI Act

NIST AI RMF

ISO/IEC 42001

Buy, build, or hybrid. Same controls. Same paper trail.

The core problem

A unit test	The same test, against an LLM
Same input →	Same input →
one deterministic output	three different outputs, three runs
Pass or fail	"Correct" isn't pass or fail

Four patterns for safe rollouts

01

Blue-Green

Fast rollback, trust your validation

02

Canary

Gradual, real-user exposure

03

Shadow

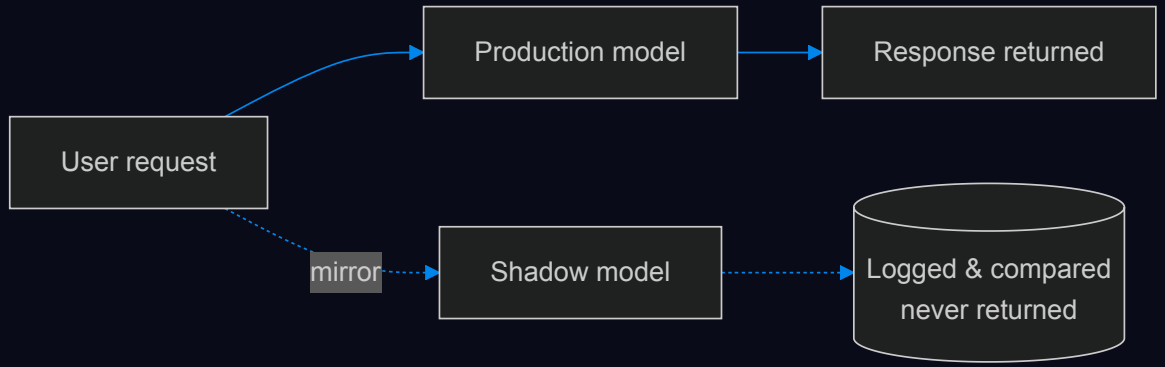
Mirror traffic, discard results

04

A/B Testing

Prove business impact

Shadow deployment, in detail



Build 1 — production path only

Build 2 — shadow duplication added

Build 3 — comparison and diff, labeled

The Azure template

01

Deploy green

0% traffic

02

Smoke test

Invoke green by name

03

Mirror

Live traffic duplicated

04

Progressive shift

10% → 25% → 50% → 100%

05

Retire blue

Trace-based eval



AgentCore Evaluations (GA June 17, 2026, AWS Summit New York) · Foundry trace-based evaluation · Agent Evaluation on Agent Platform

The sequence

01

Shadow

Tells you whether the model is broken

02

Canary

Tells you whether users tolerate it

03

Full cutover

"

If shadow isn't in your pipeline, that's your highest-leverage fix.

Why RAG exists

Foundation models hallucinate

Foundation models are frozen at training cutoff

Foundation models don’t know your company

RAG solves all three.

Four pipelines, one shape



	AWS	GCP	Azure
Collection & storage	S3	Cloud Storage	Blob Storage
Embedding & retrieval	OpenSearch	Vertex AI (Agent Platform)	Azure AI Search

The managed-knowledge shift

2024

A dev, staring at the four-pipeline diagram

Custom code, everywhere

2026

The same dev

One box: managed knowledge base

Bedrock Managed KB (AWS Summit NY, Jun 2026) · Foundry IQ (Build 2026) · GCP RAG Engine

Classic vs. agentic retrieval

Classic RAG

Query: one vector search

Sources: one

Latency: fast

Cost: cheap

Best for: FAQ, refunds, policy lookup

Agentic retrieval

Query: LLM decomposes to sub-queries

Sources: many, parallel

Latency: slow

Cost: expensive

Best for: complex, multi-source, conversational

The default has flipped

In 2026, agentic retrieval ships natively on all three clouds.

Classic RAG is still a good answer more often than teams think.

Per each platform's current product docs — Bedrock, Foundry IQ, RAG Engine

"

When the agent retrieves, who is it acting as?

"

Pick your RAG shape deliberately. Managed unless you have a real reason.

Wait — Doesn't MCP Replace RAG?

RAG is a **technique**	MCP is a **protocol**
Chunk documents	One server per resource
Embed them	Any compliant client can call it
Store in a vector database	Works across frameworks and vendors
Retrieve relevant chunks at query time	Says nothing about retrieval logic itself
Inject into the prompt	

They're not competing. MCP doesn't replace RAG — it's often how you *expose* a RAG pipeline to an agent. Your vector search doesn't disappear. It gets wrapped as an MCP server instead of hardcoded into your app.

What is an agent, actually

Model + tool	Model in a loop
Input → arrow → output	Tools, observations feeding back
Not an agent	Agent

New failure modes

Infinite loops

Wrong tool called

Hallucinated tool

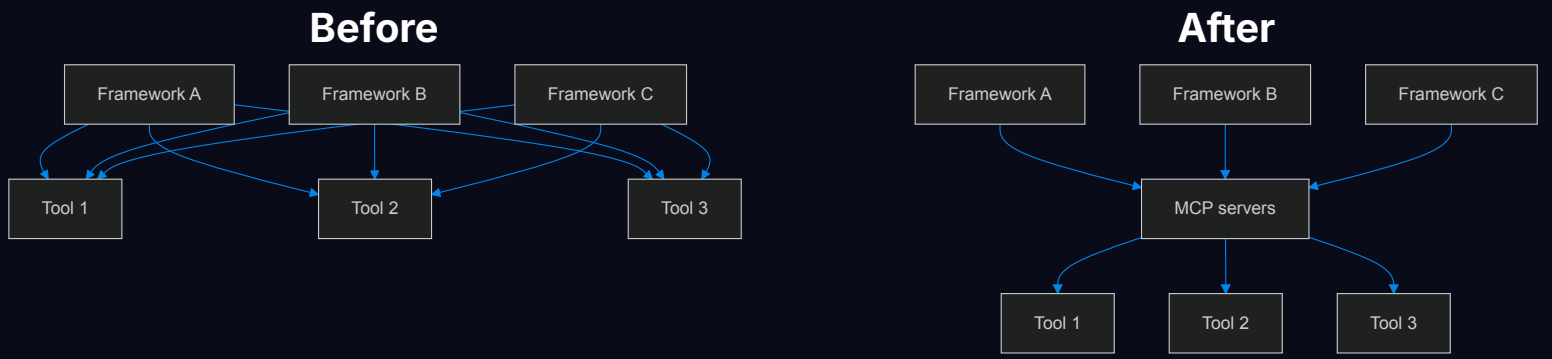
Correct tool, wrong argument

Correct result, misinterpreted

None of these existed in request/response.

Your orchestration pattern decides which of these you fight.

MCP: the setup



Model Context Protocol. Anthropic, November 2024. Open standard.

MCP: the numbers

10,000+

Public MCP servers

Anthropic's own figure, December 2025

5

Vendors (currently) shipping native support

Per each vendor's own product announcements

Dec 2025

Donated to Agentic AI Foundation

Anthropic's donation announcement, Dec 2025

July 2026

Largest spec revision since launch

Per a third-party MCP version tracker, 2026

Six orchestration patterns

Sequential	Routing	Parallel
Structured pipelines	Clear category dispatch	Latency reduction, diverse perspectives

ReAct	Hierarchical (Magentic)	Evaluator
Tool use, iterative	Open-ended tasks	Quality-critical outputs

Magentic-One: Microsoft Research

Runtime landscape, current state

AWS AgentCore

GA Oct 2025. Policy, Evaluations, and managed harness all GA June 17, 2026 (AWS Summit NY). Payments (Coinbase, Stripe) announced.

GCP Agent Platform

Announced Cloud Next '26. Vertex retired. Skill Registry. Memory Bank profiles GA.

Microsoft Foundry

Agent Framework 1.0 GA. Foundry Agent Service GA. Hosted Agents targeted GA early July 2026. M365 publishing.

AWS Summit New York, June 2026 · Google Cloud Next '26 · Microsoft Build 2026

Cross-vendor reality

Model	AWS	GCP	Azure
Claude	native, Bedrock	available, Model Garden	GA June 29, 2026
GPT	increasingly available		primary, expanding
Gemini		primary	cross-cloud emerging
Open models	everywhere	everywhere	everywhere

The frontier models have converged closely on most public benchmarks. The platform gap on governance is the more durable one.

Claude on Microsoft Foundry: Microsoft & Anthropic, June 29, 2026

"

In 2026, agents run everywhere.
What varies is who they answer to.

A2A + AP2

Two agents, different organizations, talking directly.

- **A2A** — agent-to-agent
- **AP2** — agent payments

Watch this space.

A2A and AP2: Google, open protocols

The two identity patterns

Identity passthrough

- Microsoft Foundry + Fabric
- User's Entra token flows through the agent
- Agent inherits the human

Runtime isolation

- GCP Agent Identity
- SPIFFE cert bound to container, mTLS everywhere
- Agent has its own identity, tied to where it runs

OpenTelemetry GenAI conventions

```
gen_ai.system: "anthropic"
gen_ai.request.model: "claude-opus-5"
gen_ai.usage.input_tokens: 1834
gen_ai.usage.output_tokens: 412
gen_ai.tool.name: "search_knowledge_base"
gen_ai.server.time_to_first_token: 0.312
```

Datadog · Honeycomb · Grafana · LangChain · CrewAI · AutoGen

The de facto standard. Not yet formally Stable.

As of July 2026 — OpenTelemetry GenAI semantic conventions, Development status

Cost reality

Traditional compute

Mostly flat once provisioned

Inference

Linear with usage

No cap

78% of companies run 2+ LLM families

3+ jumped from 36% to 59% in one quarter

— Databricks, State of AI Agents 2026

"

Platform choice matters more than model choice.

The nine questions

1. Are your feedback loops wired from monitoring back to framing?
2. Do you have a shadow stage before real users?
3. Classic or agentic retrieval — deliberate choice?
4. Which orchestration pattern? Can you draw it in 30 seconds?
5. Are your tools exposed via MCP?
6. Can your agent ever see data its user can't? Structurally, not policy?
7. Are you emitting OpenTelemetry GenAI conventions?
8. Article 50 enforceable 5 weeks ago — are you ready?
9. Do you know your per-inference cost and have a routing strategy?

"

The engineering already exists.
The standards are here.
The regulators are ready.
Stop dreaming.

Thank You

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